Evaluation report

How the rule engine performs on the 50-claim validation set (synthetic teaching data).

20 October 2026

Evaluation report — ClaimGuard AI (Phase 2)

Generated 2026-10-20 by `apps/api/scripts/make_eval_report.py`*. All numbers are computed, not typed.*

1. Reproducibility

- Commit: `2ee8f6e`
- Ruleset fingerprint (sha256 of rule files, first 16): `f2c3622f403ec4e4`
- Command: `cd apps/api && python scripts/make_eval_report.py`
- Rulebook: fictional teaching rulebook v1.0.0 (04_Rulebook); metrics follow 07_Evaluation_and_Acceptance.

2. Data discipline (declaration)

- Development split: used to build and tune the rule engine. **Perfect scores on it are expected and are not evidence of generalisation.**
- A commit was frozen (tag `phase2-freeze`) before any validation run.
- Validation / stress splits: reported below only if their labelled files were available in `data/evaluation/`. If validation labels influence any later change, that change must be listed in section 6.
- Held-out inputs/labels were never available to the team.

3. Summary

Split	Claims	Pairs	Macro F1	Issue P	Issue R	Issue F1	False-alarm	Status acc.	Claim exact	p50 / p95 ms
development	400	6000	1.000	1.000	1.000	1.000	0.000	1.000	400/400	0.82 / 1.61
validation	50	750	0.933	1.000	1.000	1.000	0.000	1.000	50/50	0.70 / 0.98

Metric definitions: issue = FAIL status at claim-rule level; undefined precision/recall is `null` (never 100%); an undefined per-rule F1 counts as 0 in the macro average. False-alarm rate = predicted FAIL among pairs whose expected status is not FAIL.

4. Split `development`

Missing pairs: 0 · extra pairs: 0 · false abstentions: 0 · missed abstentions: 0

Rule	TP	FP	FN	Precision	Recall	F1	Status acc.	Evidence mismatch
R001	36	0	0	1.000	1.000	1.000	1.000	0
R002	11	0	0	1.000	1.000	1.000	1.000	0
R003	28	0	0	1.000	1.000	1.000	1.000	0
R004	28	0	0	1.000	1.000	1.000	1.000	0
R005	26	0	0	1.000	1.000	1.000	1.000	0
R006	24	0	0	1.000	1.000	1.000	1.000	0
R007	23	0	0	1.000	1.000	1.000	1.000	0
R008	11	0	0	1.000	1.000	1.000	1.000	0
R009	15	0	0	1.000	1.000	1.000	1.000	0
R010	15	0	0	1.000	1.000	1.000	1.000	0
R011	10	0	0	1.000	1.000	1.000	1.000	0
R012	23	0	0	1.000	1.000	1.000	1.000	0
R013	30	0	0	1.000	1.000	1.000	1.000	0
R014	20	0	0	1.000	1.000	1.000	1.000	0
R015	19	0	0	1.000	1.000	1.000	1.000	0

Confusion (expected → predicted):

Expected	Predicted	Count
FAIL	FAIL	319
NOT_APPLICABLE	NOT_APPLICABLE	487
PASS	PASS	5014
UNABLE_TO_ASSESS	UNABLE_TO_ASSESS	180

4. Split `validation`

Missing pairs: 0 · extra pairs: 0 · false abstentions: 0 · missed abstentions: 0

Rule	TP	FP	FN	Precision	Recall	F1	Status acc.	Evidence mismatch
R001	6	0	0	1.000	1.000	1.000	1.000	0
R002	1	0	0	1.000	1.000	1.000	1.000	0
R003	3	0	0	1.000	1.000	1.000	1.000	0
R004	4	0	0	1.000	1.000	1.000	1.000	0
R005	3	0	0	1.000	1.000	1.000	1.000	0
R006	8	0	0	1.000	1.000	1.000	1.000	0
R007	4	0	0	1.000	1.000	1.000	1.000	0
R008	1	0	0	1.000	1.000	1.000	1.000	0
R009	0	0	0	null	null	null	1.000	0
R010	3	0	0	1.000	1.000	1.000	1.000	0
R011	2	0	0	1.000	1.000	1.000	1.000	0
R012	3	0	0	1.000	1.000	1.000	1.000	0
R013	5	0	0	1.000	1.000	1.000	1.000	0
R014	3	0	0	1.000	1.000	1.000	1.000	0
R015	1	0	0	1.000	1.000	1.000	1.000	0

Confusion (expected → predicted):

Expected	Predicted	Count
FAIL	FAIL	47
NOT_APPLICABLE	NOT_APPLICABLE	75
PASS	PASS	600
UNABLE_TO_ASSESS	UNABLE_TO_ASSESS	28

5. Latency

Measured in-process per claim (15 rules), single thread, on the author's machine; excludes HTTP, database and audit-log I/O. Treat as an order of magnitude.

6. Limitations (to keep honest)

- Synthetic, rule-based benchmark: high accuracy here says nothing about real payer data or clinical correctness.
- Evidence checks compare JSON pointer + observed value; they do not prove semantic relevance.
- Ambiguous inputs follow the rulebook literally (no trimming, case-sensitive IDs): e.g. a whitespace-only authorization id counts as present (R008); numeric strings in `quantity` give UNABLE_TO_ASSESS (R013). Real data may need a documented policy for these.
- Deterministic checks carry `confidence=null`; routing to human review is based on status and severity, not on a probability.
- Audit log is a local hash-chained file (tamper-evident, not immutable); anchors must be stored elsewhere.
- LLM-assisted explanations (if used) are evaluated separately; a model failure must never remove a deterministic finding.